

```
from google.colab import files
```

```
uploaded = files.upload()
```

jan to may p...b166 (1).csv

jan to may police violation_anonymized791b166 (1).csv(text/csv) - 109589874 bytes, last modified: 20/06/2026 - 100% done

```
import pandas as pd
```

```
file_name = list(uploaded.keys())[0]
```

```
df = pd.read_csv(file_name)
```

```
print("Dataset Shape:", df.shape)
```

```
df.head()
```

Dataset Shape: (298450, 24)

	id	latitude	longitude	location	vehicle_number	vehicle_type
0	FKID000000	12.925557	77.618665	18th Main Road, Block 2, Koramangala, Bengalur...	FKN00GL0000	CAR
1	FKID000001	12.905463	77.700778	Sarjapura Main Road, The Grove, Janatha Colony...	FKN00GL0001	CAR
2	FKID000002	12.925449	77.618504	Koramangala 2nd Block, Kormangala West, Bengal...	FKN00GL0002	CAR
3	FKID000003	12.956521	77.518618	6th Cross Road, Manasa Layout, Nagarbhavi, Ben...	FKN00GL0003	SCOOTER
4	FKID000004	12.977767	77.580545	Kalidasa Road, Gandhinagar, Nehru Nagar, Benga...	FKN00GL0004	TANKER

5 rows × 24 columns

```
import pandas as pd
import numpy as np

print("="*80)
print("DATASET OVERVIEW")
print("="*80)

print(f"\nRows: {df.shape[0]}")
print(f"Columns: {df.shape[1]}")

print("\nCOLUMN NAMES")
print("-"*80)
```

```
for i,col in enumerate(df.columns):  
    print(f"{i+1}. {col}")
```

=====

DATASET OVERVIEW

=====

Rows: 298450

Columns: 24

COLUMN NAMES

1. id
2. latitude
3. longitude
4. location
5. vehicle_number
6. vehicle_type
7. description
8. violation_type
9. offence_code
10. created_datetime
11. closed_datetime
12. modified_datetime
13. device_id
14. created_by_id
15. center_code
16. police_station
17. data_sent_to_scita
18. junction_name
19. action_taken_timestamp
20. data_sent_to_scita_timestamp
21. updated_vehicle_number
22. updated_vehicle_type
23. validation_status
24. validation_timestamp

```
print("\nDATA TYPES")
```

```
print("-"*80)
```

```
dtype_df = pd.DataFrame({  
    'Column': df.columns,  
    'Datatype': df.dtypes.values  
})
```

```
display(dtype_df)
```

DATA TYPES

	Column	Datatype	
0	id	object	
1	latitude	float64	
2	longitude	float64	
3	location	object	
4	vehicle_number	object	
5	vehicle_type	object	
6	description	float64	
7	violation_type	object	
8	offence_code	object	
9	created_datetime	object	
10	closed_datetime	float64	
11	modified_datetime	object	
12	device_id	object	
13	created_by_id	object	
14	center_code	float64	
15	police_station	object	
16	data_sent_to_scita	bool	
17	junction_name	object	
18	action_taken_timestamp	float64	
19	data_sent_to_scita_timestamp	object	
20	updated_vehicle_number	object	
21	updated_vehicle_type	object	
22	validation_status	object	
23	validation_timestamp	object	

Next steps: [Generate code with dtype_df](#) [New interactive sheet](#)

```
print("\nMISSING VALUES")
print("-"*80)
```

```
missing = pd.DataFrame({
    'Column': df.columns,
    'Missing Count': df.isnull().sum().values,
    'Missing %': round((df.isnull().sum()/len(df))*100,2).values
})

missing = missing.sort_values(
    by='Missing Count',
    ascending=False
)

display(missing)
```

MISSING VALUES

	Column	Missing Count	Missing %	
6	description	298450	100.00	
10	closed_datetime	298450	100.00	
18	action_taken_timestamp	298450	100.00	
19	data_sent_to_scita_timestamp	256289	85.87	
21	updated_vehicle_type	125254	41.97	
23	validation_timestamp	125254	41.97	
22	validation_status	125254	41.97	
20	updated_vehicle_number	125254	41.97	
14	center_code	11260	3.77	
3	location	3041	1.02	
15	police_station	5	0.00	
13	created_by_id	5	0.00	
17	junction_name	5	0.00	
1	latitude	0	0.00	
0	id	0	0.00	
2	longitude	0	0.00	
7	violation_type	0	0.00	
5	vehicle_type	0	0.00	
4	vehicle_number	0	0.00	
11	modified_datetime	0	0.00	
12	device_id	0	0.00	
8	offence_code	0	0.00	
9	created_datetime	0	0.00	
16	data_sent_to_scita	0	0.00	

Next steps:

Generate code with missing

New interactive sheet

```
print("\nUNIQUE VALUES PER COLUMN")
print("-"*80)
```

```
unique_df = pd.DataFrame({  
    'Column': df.columns,  
    'Unique Values': df.nunique().values  
})  
  
display(unique_df)
```

UNIQUE VALUES PER COLUMN

	Column	Unique Values	
0	id	298450	
1	latitude	177982	
2	longitude	177378	
3	location	10942	
4	vehicle_number	231890	
5	vehicle_type	22	
6	description	0	
7	violation_type	991	
8	offence_code	991	
9	created_datetime	94417	
10	closed_datetime	0	
11	modified_datetime	298450	
12	device_id	3070	
13	created_by_id	2666	
14	center_code	52	
15	police_station	54	
16	data_sent_to_scita	2	
17	junction_name	169	
18	action_taken_timestamp	0	
19	data_sent_to_scita_timestamp	42161	
20	updated_vehicle_number	143133	
21	updated_vehicle_type	22	
22	validation_status	5	
23	validation_timestamp	170115	

Next steps: [Generate code with unique_df](#) [New interactive sheet](#)

```
print("\nSAMPLE VALUES FROM EACH COLUMN")
print("-"*80)
```



```
for col in df.columns:
    print(f"\n{col}")
    print(df[col].dropna().unique()[:5])
```

```
offence_code
['[112,104]' '[113]' '[112,107]' '[112]' '[112,113,111,109,108,107]']
```

```
created_datetime
['2023-11-20 00:28:46+00' '2023-11-24 22:46:46+00'
 '2023-11-20 00:27:46+00' '2023-11-16 06:47:46+00'
 '2023-11-22 04:56:46+00']
```

```
closed_datetime
[]
```

```
modified_datetime
['2023-11-28 04:48:04.582978+00' '2023-11-24 23:00:24.115257+00'
 '2023-11-28 04:47:02.33776+00' '2023-11-18 04:46:57.216868+00'
 '2023-11-28 02:44:50.46737+00']
```

```
validation_status
['approved' 'rejected' 'processing' 'duplicate' 'created1']

validation_timestamp
['2023-11-30 03:08:24.818+00' '2023-11-30 03:08:56.998+00'
 '2023-11-18 23:35:12.428+00' '2023-11-30 03:11:32.796+00'
 '2023-11-30 01:27:10.711+00']
```

```
num_cols = df.select_dtypes(
    include=['int64', 'float64']
).columns
```

```
print("NUMERICAL COLUMNS")
print(num_cols.tolist())
```

```
NUMERICAL COLUMNS
['latitude', 'longitude', 'description', 'closed_datetime', 'center_code', 'acti
```

```
cat_cols = df.select_dtypes(
    include=['object']
).columns
```

```
print("CATEGORICAL COLUMNS")
print(cat_cols.tolist())
```

```
CATEGORICAL COLUMNS
['id', 'location', 'vehicle_number', 'vehicle_type', 'violation_type', 'offence_
```

```
print("NUMERICAL STATISTICS")
```

```
display(
    df.describe().T
)
```

NUMERICAL STATISTICS							
	count	mean	std	min	25%	75%	max
latitude	298450.0	12.980802	0.049732	12.802667	12.963331	12.977500	12.990000
longitude	298450.0	77.600512	0.050518	77.442553	77.571198	77.580000	77.590000
description	0.0	NaN	NaN	NaN	NaN	NaN	NaN
closed_datetime	0.0	NaN	NaN	NaN	NaN	NaN	NaN
center_code	287190.0	23.023013	20.006118	2.000000	11.000000	17.000000	20.000000
action_taken_timestamp	0.0	NaN	NaN	NaN	NaN	NaN	NaN

```
for col in cat_cols:

    print("\n"+"="*80)
    print(col)

    display(
        df[col]
        .value_counts(dropna=False)
        .head(10)
    )
```